

Suggested FYP topics for 2026-27

by

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Project code :	XYY1
Project title :	An Accurate and Efficient Machine Unlearning System for Harmful Content Removal in Multimodal Large Language Models
No. of students :	1
Description of project :	MLLMs have achieved remarkable success in multimodal tasks and are becoming increasingly popular. However, their rapid development has also brought critical challenges, including safety risks, bias in content generation, privacy leakage, and copyright violations. Machine unlearning offers a promising approach to addressing these issues by enabling models to selectively remove harmful, outdated, or sensitive knowledge. However, ensuring effective unlearning while preserving model utility remains a significant challenge. Moreover, as MLLMs are still in the early stages of commercialization and large-scale deployment, many of their harmful behaviors remain insufficiently explored. Meanwhile, AI-related laws and regulations are rapidly evolving, requiring models to continually adapt to meet new societal and legal standards. This project encourages students to explore existing methods in both continual learning and machine unlearning, with a focus on adapting them to MLLMs. The goal is to develop an efficient unlearning method for MLLMs that enables the selective forgetting of targeted visual concepts while minimizing data requirements. Additionally, the method should preserve general model capabilities, ensuring that only specified knowledge is removed without degrading overall performance.
Industrial collaboration :	No.

Project code :	XYY2
Project title :	Sketch2Art: A Conditional Diffusion Model for Creative Visual Generation
No. of students :	2
Description of project :	This project employs a conditional diffusion model to generate imaginative visuals, ranging from cartoons and animations to painterly illustrations, through user-provided sketches and text prompts. By learning from diverse data, the model adeptly integrates structural cues and semantic guidance, producing high-quality, coherent images that capture each user's unique creative vision. The final deliverable is an interactive demo in which users can experiment with a broad range of inputs, from rough sketches to detailed annotations, and receive synthesized results. This flexible setup accommodates a wide variety of artistic possibilities, making it an intuitive, practical tool for creative

	expression.
Industrial collaboration :	No.

Project code :	XY Y3
Project title :	Building a Robust Safeguard for Vision-language Models
No. of students :	2
Description of project :	Vision-Language Models (VLMs) serve as foundational components for multimodal intelligence, yet their susceptibility to adversarial visual inputs poses significant safety challenges in real-world applications. While textual safeguards are relatively effective, visual adversarial attacks can easily bypass existing defenses, leading to unsafe or unintended outputs. Current defense mechanisms are either computationally expensive, requiring extensive data and retraining, or struggle to balance safety with response usefulness. To address these challenges, this project aims to develop a novel inference-time alignment framework that enhances VLM safety without compromising utility. On the one hand, both input visual content and generated responses to establish robust multimodal safety awareness. On the other hand, VLM outputs are aligned by conditioning the generative distribution with an interference prefix and employing sentence-level best-of-N search to select the most harmless yet helpful response. This safeguard framework ensures effective safety alignment while maintaining the efficiency and adaptability needed for real-world deployment.
Industrial collaboration :	No.

Project code :	XY Y4
Project title :	A Literature Survey System Based on LLM Agent
No. of students :	2
Description of project :	Develop an LLM (Large Language Model)-based agent system to achieve fully automated literature retrieval, key information extraction, and trend analysis, helping researchers quickly grasp the dynamics of a field. Motivation: - The exponential growth of academic publications (over 5 million papers published annually in STEM fields alone) has created critical challenges for researchers. Researchers struggle to identify high-impact papers among massive publications. Autonomous agents can address this problem by simulating human researcher behaviors (e.g., iterative query refinement). Goal : The system should include the following core functions: Intelligent Retrieval: Automatically generate diverse search strategies based on user input keywords or questions. Multi-source Data Integration: Compatible with academic platforms such as PubMed, Google Scholar, and local literature databases. In-depth Analysis: Extract structured information such as

	research content, proposed method, experiment analysis, and main conclusion. 4. Visual Report Generation: Output a PDF document that includes a research summary, timeline analyses, and suggestions for research gaps. Higher Goal • Visualization: Gradio/PyQt Interface • The output report includes various figures and tables. Reference • https://github.com/mannaandpoem/OpenManus • https://github.com/geekan/MetaGPT • https://manus.im/ • Agent-as-a-Judge: Evaluate Agents with Agents
Industrial collaboration :	No.

Project code :	XY55
Project title :	<Any Other Interesting Topics that You Like>
No. of students :	1 or 2 or 3
Description of project :	Please send me an email and we can further discuss your idea and interest.
Industrial collaboration :	No.

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